Oleg Presnyakov

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SUMMARY

Mathematics major with a strong background in competitive programming, algorithm design, and computational biotech research. Experienced in developing custom bioinformatics algorithms, image-labeling software, and machine-learning pipelines for antibody research. Contributed to research publications and improved baseline model performance by 18%.

WORKING EXPERIENCE

Avery (RIPS) Researcher **Los Angeles (UCLA), CA**
Math Researcher *June 2025–Present (Full time after internship)*

- Integrating advanced machine learning with physics-informed modeling to streamline protein engineering processes.
- Evaluated and benchmarked multiple sequence-analysis tools across key performance metrics, identifying the optimal solution for protein engineering applications.
- Integrated selected software into unified Bayesian optimization pipeline and achieved an 18% improvement in key experimental performance metrics.
- Developed two custom DNA-cutting algorithms for optimal reconstruction on the company's product chip and nanochannel platforms; validated on GFP (chip) and *E. coli* (1M bases, nanochannel).
- Designed a droplet-labeling platform using PCA and iterative blurring, accelerating labeling throughput by 20×.
- Deployed a local LLM stack on EXO with a custom internal chat interface, and built supporting scientist tools including a lab notebook and inventory system.

`https://arxiv.org/pdf/2510.24974`

UCI Math **Irvine, CA**
Numerical Analysis Research *September 2024–June 2025*

- Optimized the Kaczmarz algorithm by introducing a novel pairwise-comparison strategy, enhancing robustness when interacting with users and handling noisy data.
- Designed clustering-based methods to generate compact coresets for overdetermined linear systems efficiently. (`https://github.com/janguyen115/Approaches-to-Randomized-Kaczmarz-for-Ill-Conditioned-Systems-`)

AI Grader **Irvine, CA**
AI grading system for math lower-division classes *December 2024–April 2025*

- Integrated ChatGPT API requests to automate grading workflows for lower-division math courses.
- Leveraged LLM prompting to auto-generate and refine grading rubrics, optimizing criteria extraction.
- Cleaned and curated UCI student response data, employing self-labeling techniques to produce training scores for prompt-based grading.

EDUCATION

University of Michigan **Ann Arbor, MI**
Phd in Electrical and Computer Engineering *September 2026*

- Incoming student

UNIVERSITY OF CALIFORNIA, IRVINE **Irvine, CA**
Bachelor of Science in Mathematics *September 2023 - December 2025*

- Completed all requirements for both Bachelor's and Master's degrees within 3 years.
- GPA: 3.83; Dean's Honor List since Winter 2023.

PROJECTS

- ICPC SoCal Regionals 2024–2025: 8th Place (Bronze)
- Codeforces (`FANAT_MORGENSHTERNA`): `https://codeforces.com/profile/FANAT_MORGENSHTERNA`
- UCI Datathon 2025: Pioneered a hybrid clustering–Held-Karp approach to efficiently approximate TSP tours on 350 nodes (`https://devpost.com/software/dronedash-santa-s-test-flight-at-uci`)

SKILLS

- **Programming:** Python, C++
- **ML & Optimization:** PyTorch, BoTorch, TensorFlow, Deep learning, Bayesian optimization
- **Bioinformatics & Data Analysis:** AlphaFold/ColabFold, HADDOCK3, NumPy, pandas, GROMACS
- **Tools & Collaboration:** Git, SSH, SSHFS, Google Cloud VM, problem-solving